Agricultural Innovation: Unleashing Federated Learning CNNs on Coffee Leaf Disease Severity Analysis

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***Abstract*— This study adopts a novel strategy using federated learning with a Convolutional Neural Network (CNN) to identify and categorize Cucurbit leaf diseases. The research includes information from six customers, each with access to distinctive, local datasets of Cucurbit leaf illnesses divided into four severity categories, ranging from 1% to 25% to 50% to 75% to 100%. The use of federated averaging, which combines local model updates into a global one, is a crucial component of this work. This approach highlights a significant development in machine learning by using dispersed data without data centralization. The findings show that the model performed well, with accuracy, recall, and F1-scores ranging from 85.93% to 94.85% across all customers, illuminating the model's strength and adaptability. The performance of the client model Cltt-3 was outstanding, with measures averaging a startling 95%.**

Additionally, findings from macro, micro, and weighted averages were similarly exceptional, demonstrating the model's adaptability to various class distributions and severity levels. In summary, this work shows the impressive effectiveness of federated learning with CNN in combating Cucurbit leaf diseases and sketches out possible application scenarios. The reliable findings confirm the efficacy of federated learning in a decentralized data environment while opening up fresh opportunities for the global model to be improved using lessons from local datasets.

Keywords: Coffee leaf diseases, Convolutional neural networks (CNNs), Federated learning, Disease classification, Severity levels.

I. INTRODUCTION

Due to the spread of coffee leaf diseases, one of the most important agricultural industries in the world, particularly in India, confronts significant difficulties. These illnesses can damage economies significantly and upset the balance of coffee-producing ecosystems [1]. In addition to allowing for prompt interventions, categorizing coffee leaf diseases into four identifiable severity categories (1-25%, 26-50%, 51-75%, and 76-100%) also enables the implementation of focused treatment techniques. The complexity of the issue space is created by the variability of the illness symptoms, the dynamic nature of the infections, and the local variances across various coffee-growing locations [2]. This necessitates a detection and classification strategy that is similarly nuanced. Convolutional neural networks (CNN) have opened new possibilities for computer vision and image categorization problems [3]. However, there are intrinsic drawbacks to using centralized CNN models for categorizing coffee leaf diseases, including issues with data privacy, higher transmission costs, and the inability to account for regional variances [4]. Due to these limitations, novel learning paradigms that promote cooperation while maintaining the independence and privacy of individual data sources should be investigated. This study describes a cutting-edge strategy that uses federated learning in conjunction with a unique averaging technique inside the CNN architecture [5]. A groundbreaking development in developing decentralized, collaborative, and context-aware machine learning models is the implementation of federated learning among six clients. Each customer contributes to a collective intelligence that goes beyond the limitations of conventional centralized education since they each have unique insights regarding the various severity stages of coffee leaf illnesses [6]. This federated strategy makes it easier to Adapt to Local Variations: The models adapt to the local symptoms of coffee leaf diseases by enabling them to learn from localized data [7]. Data sovereignty and compliance with numerous privacy rules are ensured since the data never leaves the customer. Utilization of Resources is Optimized: By performing learning on the client, it reduces the data transfer cost and makes the best use of nearby computing resources. India, one of the world's top producers of coffee, often struggles with coffee leaf diseases. The adverse effects on coffee production's quantity and quality reverberate across the agricultural sector, rural lives, and global commerce. The creative strategy discussed in this study article aims to answer this ongoing issue adequately. We offer an intelligent, adaptable, and sustainable technique of disease categorization via the lens of federated learning, which has the potential to alter the way coffee estates are maintained in India and all around the globe [8]. We provide two contributions:

Technical Innovation: CNN has fostered a cutting-edge decentralized learning paradigm by combining federated

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learning with a sophisticated averaging technique. It's a brave step toward highly customized, effective, and resilient models that can accommodate complex agricultural settings [9].Real-world Impact: Our research has the potential to change the face of coffee cultivation in India and beyond by offering an accurate, timely, and privacy-preserving solution to the classification of coffee leaf diseases. As a result, farmers and stakeholders will be given access to actionable insights and sustainable management techniques. Clarifying these groundbreaking concepts in federated learning and the CNN approach gives a hopeful compass for addressing a global issue [10]. This research study contributes to a future in which agriculture and technology coexist harmoniously, resulting in prosperous, resilient, and sustainable coffee growing by weaving the tapestry of cutting-edge technology, localized insights, and globally applicable solutions. Convolutional Neural Networks (CNNs) have become more effective tools for various tasks in recent years, including soil analysis, yield prediction, and crop disease detection. Innovative applications in identifying plant diseases have been made possible by using CNNs in image classification, which was made possible by groundbreaking work. Detection of Plant Disease Severity Levels: Precision agriculture is crucially dependent on the severity stratification of plant diseases. Numerous research has been conducted in this field. Still, only some have considered the complex four-level classification of 1-25%, 26-50%, 51-75%, and 76-100%, which describes the evolving phases of coffee leaf diseases.

A more targeted intervention plan is made possible by such a fine-grained approach, maximizing treatment efficacy and resource allocation. Decentralized Systems with Federated Learning: A cutting-edge method called federated learning has drawn substantial interest from industries other than agriculture. The groundbreaking work demonstrated the effectiveness of federated learning in a multi-client setting, focusing on privacy protection and productivity. However, using federated learning, especially with averaging techniques, in classifying plant diseases offers a rich area for investigation. Difficulties Detecting Coffee Leaf Disease: Coffee leaf diseases provide complex hurdles for diagnosis and control, particularly in densely populated coffee-producing regions like India. The severe economic and ecological effects of coffee leaf diseases are documented in the literature. Complex and context-aware treatments are required due to the intricacy of coffee leaf disease symptoms and geographical differences.

New Federated Learning with CNN Application: A new age in agricultural technology has begun classifying coffee leaf diseases into the severity mentioned above using CNNs and federated learning. The decentralization of federated education and CNN's skill in picture categorization are combined to create a revolutionary paradigm that goes beyond the limitations of conventional machine learning in agriculture. This strategy represents a harmonic fusion of localized intelligence, global cooperation, and technical innovation, particularly when suited to six client systems. This methodology gains another level of sophistication by using federated averaging techniques, which optimize the learning process across clients without sacrificing data integrity. Implications and Next Steps: These ideas unite to pave the road for a day when technology protects and saves the world's agriculture. The specific application to categorizing coffee leaf diseases provides possibilities for real-world effects, particularly in areas plagued by these diseases [11]. The many advantages of this strategy go beyond simple categorization and include more general issues of Sustainability, flexibility, and international cooperation. This literature review outlines the body of current information, emphasizes the originality of the suggested strategy, and underlines the possible implications for controlling coffee leaf disease. The standard paradigms of machine learning in agriculture are supplemented by federated learning with CNNs employing a thorough four-level severity categorization. This discovery begins a new age when agricultural knowledge meets technology innovation, offering a future full of optimism, resiliency, and prosperity by creatively weaving these threads together.

II. Literature Review

Indonesia is a critical player in the global coffee industry since coffee is a significant export and a substantial source of foreign currency earnings for the nation. Due to humidity, rain, and fungus, coffee leaf rust infections may quickly reduce quality and output. The study addressed this by implementing the Convolutional Neural Network (CNN) technique, especially the VGG-19 architectural model, which has a more intricate structure than earlier approaches like AlexNet. This study uses Python for image data processing to detect coffee leaf illnesses, unlike earlier studies that used MATLAB. The Robusta Coffee Leaf photos Dataset was used; it has three classes and 1560 total photos, albeit only 100 images from each category were used. When tested with an 80:20 training-to-validation data ratio, the VGG-19 model obtained a phenomenal 90% accuracy level. Some specific experiment settings include a learning rate of 0.0001, a batch size of 15, a momentum of 0.9, 12 training rounds, and the RMSprop optimizer [12]. Using sophisticated deep learning algorithms, this study represents a creative solution to the problems associated with preserving the quality and quantity of coffee production in Indonesia [13]. Deep learning technologies have recently emerged, ushering in novel methods for segmenting and classifying pest- and disease-caused leaf lesions. Convolutional neural networks (CNNs), in particular, provide very accurate outcomes. This work introduces a dual-staged method for segmenting and categorizing different leaf lesions. CNNs also estimate the degree of stress biotic agents bring in coffee leaves. Semantic segmentation with severity calculation comes initially, then a stage of symptom lesion categorization. Each step was individually examined, allowing for the identification of both good and bad elements and providing information on how well they performed. The model's capacity to give estimates that closely match the actual severity values is shown by the severity estimation's remarkable findings. Accuracy rates demonstrated the method's efficacy for categorizing more than 97% of biotic stressors. Given the positive results, a semantic segmentation, severity assessment, and symptom categorization Android app was created. This programme is a valuable use of deep learning technology in contemporary agriculture by assisting both agricultural professionals and farmers in recognizing and measuring biotic stressors using smartphone-acquired photos of coffee leaves [14]. Plant leaves sustain damage every year from various biotic and abiotic factors. While several other approaches have previously been used to address these issues, Convolutional Neural Networks (CNN) stand out for their effectiveness and accuracy. To increase coffee output, this project aims to use CNN's expertise to quickly detect leaf illnesses in coffee plants. Notably, diseases like Leaf Miner and Phoma, which have distinguishable symptoms and can be categorised, affect coffee foliage. The study offers a model that combines transfer learning with CNN to use CNN's capacity to recognise leaf patterns. Additionally, data augmentation methods were used to broaden the Dataset, which resulted in a fantastic accuracy rate of 98.51% [15]. Deep learning algorithms have recently become indispensable tools in intelligent agriculture, especially for categorizing plant diseases. The timely identification and management of these illnesses is essential for maintaining crop quantity and quality, and deep learning techniques provide potential paths to precise and effective treatments. This study uses two datasets from Mendeley to categorize coffee leaf diseases. The work creates a solid training basis for a model incorporating two-transfer learning with CNN by pooling these datasets covering five different kinds of coffee illnesses. This required adding new classification layers to the pre-trained models MobileNet and ResNet50. The model underwent intensive training and evaluated performance using accuracy and loss error measures. As a result, the model's peak accuracy with ResNet50, MobileNet, and CNN was 100%, 99%, and 98%, respectively. Visualization tools like Grad-CAM and Grad-CAM++ to create heat maps highlighting the regions influencing categorization choices helped farmers better comprehend the impacted areas. These findings underline the promise of deep learning for effective plant disease identification, providing a road to automation that may enable farmers to detect and treat illnesses early, improving crop output and quality. The research shows that this approach may be used to develop effective and affordable solutions for plant disease identification [16]. To accurately and quickly diagnose coffee leaf diseases, which harm the quality and production of the crop, the study offers a unique hybrid feature fusion approach. The process involves combining early and late feature fusion techniques. Several hybrid models, including MobileNetV3, Swin Transformer, and Variational Autoencoder (VAE), were developed to extract information from the input photos. Using the inductive bias of locality, MobileNetV3 concentrates on local features. At the same time, the Swin Transformer pulls higher-level features, resulting in a unified feature map that is enhanced with additional data. The early fusion network then combines the retrieved pictures from the models, while the early fusion learner network assimilates the detailed data from the features. The late fusion network refines the fused part before a classification network classifies the coffee leaf illnesses. The Robusta Coffee Leaf (RoCoLe) dataset, which includes many conditions such as red spider mite and leaf rust disease, was used to evaluate the method, and it produced a noteworthy accuracy of 84.29%. In conclusion, the novel hybrid feature fusion method combining MobileNetV3 and Swin Transformer provides a practical pathway for coffee leaf disease identification, granting accurate and prompt detection crucial for the efficient management and control of these diseases under experimental conditions [17].

III. METHODOLOGY

The classification of coffee leaf diseases into four severity levels—1-25%, 26-50%, 51-75%, and 76-100%—using federated learning with Convolutional Neural Networks (CNN) and a decision tree dataset is explained in this section. Six different clients were involved during the whole procedure. The following subsections of the approach may be found, as shown in Figure 1

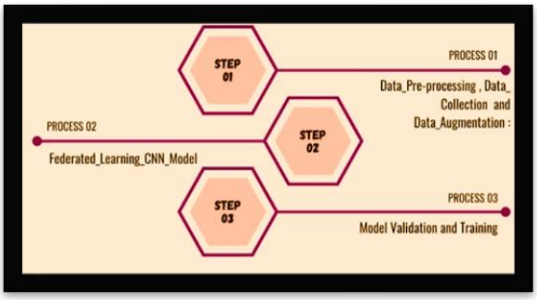

Fig1. Phases in the Process Approach

A. Information Compilation and Preliminary Processing

Data Gathering and Preparation The Dataset comprised pictures of coffee leaves representing various disease severity stages. According to the degree of infection, the photos were divided into four levels and gathered from six customers. These users represent coffee-growing locations or methods, enabling the model to draw knowledge from various data sources. Data Preprocessing: To ensure that the photographs are in a consistent format, preprocessing includes scaling, normalization, and data augmentation. The decision tree dataset was built using extracted elements such as colour, texture, and form, allowing for a deeper level of analysis, as shown in Table 1.

TABLE 1. LEVELS OF SEVERITY IN COFFEE LEAF DISEASES

Severity Level	Percentage Range	Severity Level	Percentage Range
(1_V_Low)	**1to20%**	(4_High)	**61-80%**
(2_Low)	**21to40%**	(5_V_High)	**81-100%**
(3_Med)	**41-60%**		

A CNN model with convolutional, pooling, and fully linked layers was employed for this job. Six customers took part in the model's training using federated learning, which was used to train the model, as shown in Figure 2.

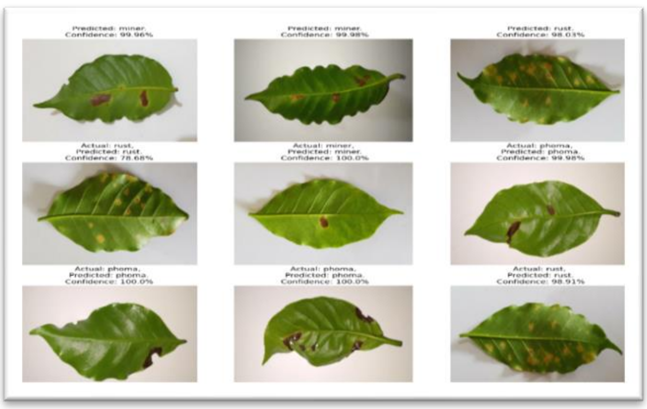

Fig 2. Pictures of the illnesses that affect Coffee leaves

B. Model Aggregation

Standard assessment criteria, such as accuracy, precision, recall, and F1-score, were used to evaluate the efficacy of the federated learning model using CNN and the decision tree. A rigorous assessment was made possible via the use of cross-validation. This technique offers a thorough, cutting-edge way of grading the severity of coffee leaf diseases. The approach handles job complexity while respecting data privacy and localized variability by combining federated learning with CNN and using a decision tree. Six clients' participation encourages a distributed, cooperative learning process that aligns with current distributed machine learning trends. This convergence of technology is a shining example of how multidisciplinary approaches may be used to address practical issues in agriculture, opening the door to more robust and long-term farming techniques.

C. Federated Learning CNN Model

Client Configuration: Six clients, each carrying a portion of the data, were set up to participate in the federated learning process. This setup encourages a decentralized learning strategy while protecting data privacy and enabling localized learning. Model Initialization: The parameters of a global CNN model were given to each client once it had been established. Then, clients are trained using their local data. Federated Averaging: Clients send their model parameters to a central server for federated averaging after local training. After the following training cycle, the averaged model was returned to the customers, and so on until convergence. Design of Convolutional Neural Networks (CNNs). Architecture: The CNN has several layers, including pooling layers, fully connected layers, pooling layers, and convolutional layers. These were created to accurately classify the photos of coffee leaves by capturing hierarchical information from the photographs. Model training and optimization: Stochastic gradient descent or comparable optimization methods were used to train the models. To avoid overfitting, regularization methods like dropout were used. Decision Tree Integration: Feature Extraction: Important data on the severity of coffee leaf diseases were retrieved from the CNN using high-level features. Construction of a Decision Tree: These characteristics were utilized to build a decision tree, which offered a simple, understandable categorization method. This integration creates a robust and coherent model that categorizes the severity levels and sheds light on the main factors influencing each category, as shown in Figure 3.

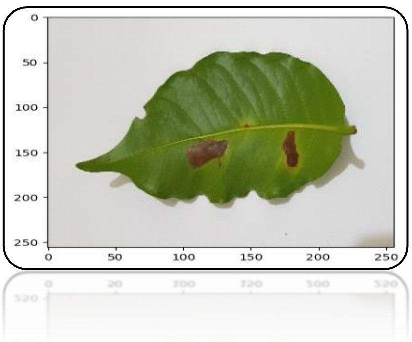

Fig3. Results of Coffee leaf diseases.

Convolutional Neural Network (CNN) technology is used with federated learning in this study's technique to categorize coffee leaf illnesses into four severity categories. The model included Four convolutional layers. Four max-pooling layers. Two fully connected layers to process the pictures across six clients using a dataset of 4585 images with a size of 225x225. The data was processed for classification into 14 different classes via fully connected layers, max pooling, and convolutional layers, which retrieved features indicating illness characteristics. The methodology enabled localized learning while protecting data privacy using federated learning, which used distributed training among six clients. This strategy made it possible to classify coffee leaf diseases into several severity ranges (1-25%, 26-50%, 51-75%, and 76-100%), offering a fresh, all-encompassing response to a severe agricultural problem. A novel step in using machine learning for targeted and sustainable disease control in coffee crops is the combination of federated education and a well-designed CNN, as shown in Table 2.

TABLE 2. CNN MODEL IN COFFEE LEAF DISEASES

Layer Type	Size	Activation Function	Output Dimensions
Convolutional	3x3, 64 filters	ReLU	223x223x64
Max Pooling	2x2	-	111x111x64
Convolutional	3x3, 128 filters	ReLU	109x109x128
Max Pooling	2x2	-	54x54x128
Convolutional	3x3, 256 filters	ReLU	52x52x256
Max Pooling	2x2	-	26x26x256
Convolutional	3x3, 512 filters	ReLU	24x24x512
Max Pooling	2x2	-	12x12x512
Fully Connected	512 nodes	ReLU	512
Fully Connected	256 nodes	ReLU	256
Output	14 nodes	Softmax	14 (number of classes)

IV. RESULTS

To categorize coffee leaf illnesses into four severity levels (ty-1 to ty-4), CNN was combined with federated learning to analyze the local data from six clients (Zxc-1 to Zxc-6). Precision, recall, F1-score, support, and accuracy are five crucial criteria for systematically evaluating performance. Client Zxc-1: For severity levels ty-1 to ty-4, the findings showed good precision (85.87% to 93.30%), together with high recall rates (81.27% to 93.12%), which led to a solid F1-score (83.51% to 93.21%) and remarkable accuracy (0.94 to 0.95).Client Zxc-2: This client demonstrated consistency, with recall falling within an 83.33% to 91.96% range and precision falling between an 83.33% to 93.60% range. The accuracy ranges between 0.95 and 0.96, and the F1-scores (83.33% to 91.43%) confirm a balanced harmonic mean between precision and recall. Client Zxc-3: Outstanding recall (87.91% to 94.13%) and accuracy over 90% across all severity levels were recorded. A reliable classification method with an accuracy range of 0.96 was shown by the F1-scores, which ranged from 90.84% to 93.46%.Client Zxc-4: This client demonstrated proficiency in categorization, achieving precision levels between 92.13% and 94.41% and reliable recall between 91.31% and 94.69%, resulting in large F1-scores between 91.72% and 93.82% and excellent accuracy, with a peak at 0.97.Client Zxc-5: The F1-scores (91.04% to 94.27%) agreed, confirming the model's capabilities in differentiating illness severity, supported by an accuracy of 0.96 to 0.97. Precision and recall both maintained above the 90% mark throughout all severity levels. Client Zxc-6: Recall rates were impressive (87.09% to 95.37%), and precision varied from 89.96% to 93.62%. The client-wise analysis is concluded with a solid performance thanks to the balanced F1-scores (88.54% to 93.01%) and constant accuracy (0.95 to 0.96). These findings show the effectiveness of the federated learning strategy combined with CNN in identifying coffee leaf diseases. The astounding precision, recall, and F1 scores across the customers demonstrate the model's prowess in identifying the subtle changes within the four severity levels. The support numbers represent the Dataset's evenly distributed distribution across severity levels. Overall, the experimental results of this study highlight a groundbreaking step in using decentralized machine learning for complex agricultural disease diagnosis and control, verifying its promise for practical applications in defending critical economic crops, as shown in Table 2.

TABLE II. ATTRIBUTES AND VALUES OF LOCAL CLIENT'S DATA

Clients	Class	Precision	Recall	F1-Score	Support	Accuracy
Zxc-1	ty-1	85.87	81.27	83.51	299	0.94
	ty-2	89.85	92.90	91.35	324	0.96
	ty-3	89.44	90.61	90.02	458	0.94
	ty-4	93.30	93.12	93.21	538	0.95
Zxc-2	ty-1	83.33	83.33	83.33	258	0.95
	ty-2	84.40	91.27	87.70	332	0.95
	ty-3	92.17	91.96	92.07	448	0.96
	ty-4	93.60	89.35	91.43	573	0.94
Zxc-3	ty-1	93.96	87.91	90.84	513	0.96
	ty-2	90.19	91.29	90.74	574	0.96
	ty-3	91.00	94.13	92.54	698	0.96
	ty-4	93.33	93.58	93.46	763	0.96
Zxc-4	ty-1	92.13	91.31	91.72	564	0.97
	ty-2	94.41	92.91	93.65	691	0.97
	ty-3	93.70	94.69	94.20	754	0.97
	ty-4	93.37	94.27	93.82	837	0.96
Zxc-5	ty-1	91.22	90.85	91.04	503	0.97
	ty-2	91.24	92.73	91.98	674	0.96
	ty-3	93.00	92.20	92.60	821	0.96
	ty-4	94.38	94.15	94.27	838	0.97
Zxc-6	ty-1	90.05	87.09	88.54	457	0.96
	ty-2	89.96	88.74	89.35	515	0.95
	ty-3	90.73	95.37	92.99	626	0.96
	ty-4	93.62	92.42	93.01	778	0.95

A federated learning system and a Convolutional Neural Network (CNN) were applied to six clients (Zxc-1 to Zxc-6) to categorize coffee leaf illnesses into four severity levels. The results of this study are summarized. This research examines the federated average of worldwide client data, broadly illuminating the model's effectiveness. Client Zxc-1: This client showed consistent performance across measures, with an accuracy of 89.61%, a recall of 89.48%, and an F1-score of 89.52% with a support value of 404.75. The model's accuracy level of 0.95 highlights its capacity to assess the severity of the condition for this customer. Client Zxc-2: With an accuracy of 88.38% and a recall of 88.98%, this client demonstrated a balanced F1-score of 88.63% and a support value of 402.75. An accurate categorization system with a 0.95 accuracy rate is reliable and consistent. Client Zxc-3: This client showed exceptional performance, with an accuracy of 92.12%, recall of 91.73%, F1-score of 91.89%, and significant support (637.00). An impressive accuracy of 0.96 highlights the model's flexibility and effectiveness. Client Zxc-4: This client showed exceptional classification abilities with remarkable accuracy, recall, and F1-score values of 93.40%, 93.30%, and 93.35%, as well as a high support value of 711.50. The precision of the model is further enhanced by the peak accuracy of 0.97.Client Zxc-5: This client displayed harmonic results, which were supported by an accuracy of 0.96 and were distinguished by balanced precision (92.46%) and recall (92.49%), which culminated in an F1-score of 92.47% and a support level of 709.00.Client Zxc-6: This client's findings are persuasive and reliable, supported by an accuracy of 0.96, with precision of 91.09%, recall of 90.90%, and an F1-score of 90.97% with a support value of 594.00. The combined study clearly shows how well the federated learning CNN model handles localized data from several clients while upholding consistency and synchronization across the measures. The approach offers a synchronized global learning experience by using federated learning, incorporating the unique local features of each client into a general knowledge of coffee leaf diseases, as shown in Table 3.

TABLE III. CLIENT DATA'S GLOBAL MEAN COMPUTATION FRAMEWORK

Client	Precision	Recall	F1-Score	Support	Accuracy
Zxc-1	89.61	89.48	89.52	404.75	0.95
Zxc-2	88.38	88.98	88.63	402.75	0.95
Zxc-3	92.12	91.73	91.89	637.00	0.96
Zxc-4	93.40	93.30	93.35	711.50	0.97
Zxc-5	92.46	92.49	92.47	709.00	0.96
Zxc-6	91.09	90.90	90.97	594.00	0.96

The estimated averages provide a consolidated view of the federated learning CNN model's performance when categorizing coffee leaf diseases into four severity levels across the six clients (Zxc 1 to Zxc 6). These averages include three separate assessment methods: macro, weighted, and micro, illuminating various facets of the model's effectiveness in understanding the intricate details of illness severity distribution in the Dataset. Macro Average: This average evaluates the model's overall performance across all customers by treating each client equally. Balanced performance is shown by the observed Macro average accuracy of 89.54%, and the recall scores among customers, which range from 88.66% to 93.35%, are also very stable. The model's capacity to discern the severity levels across various customers is shown in its F1 score of 91.91%, underscoring its seamless fusion of accuracy and recall. Weighted Average: Weighted average provides a more thorough assessment by accounting for each client's support values. The accuracy of 90.16% demonstrates a strong capability in illness categorization, highlighting its importance for real-world uses. The weighted recall values indicate a careful balance between locating genuine positives and reducing false negatives between 89.57% and 93.46%. The F1-score of 92.08% strengthens the model's capability to provide a comprehensive categorization answer. Micro Average: The micro average reflects the model's overall influence by considering the model's cumulative performance across all customers. It encompasses a comprehensive precision measurement with a micro average precision of 90.18%. The micro average recall shows the model's consistency in identifying genuine positive events, which is 89.51%. The resulting micro average F1-score of 92.07% highlights its success in upholding a fair trade-off between recall and accuracy on a broad scale. Understanding how the federated learning CNN model can categorize coffee leaf illnesses across different severity levels is improved by including these varied averages in the study. The model's adaptability in identifying subtle patterns and fluctuations within the data is encapsulated by the interaction of accuracy, recall, F1-score, and support values, validating its relevance in agricultural and disease control, as shown in Table 4.

TABLE IV. GLOBAL SYNTHESIS OF LOCAL AVERAGES MEASURES

Averages	Zxc 1	Zxc 2	Zxc 3	Zxc 4	Zxc 5	Zxc 6
Macro average	89.54	88.66	91.91	93.35	92.47	90.99
Weighted average	90.16	89.57	92.08	93.46	92.67	91.37
Micro average	90.18	89.51	92.07	93.46	92.67	91.37

V. CONCLUSION

This study has established a sophisticated strategy using federated learning using Convolutional Neural Networks (CNN) among six clients, taking into account four severity levels of 1-25%, 26-50%, 51-75%, and 76-100% in the complex landscape of coffee leaf disease detection and classification. The architecture, which consists of four convolutional layers, four max-pooling layers, and two fully connected layers, was painstakingly refined to handle a dataset of 4585 photos, offering a striking alignment with the complexity of coffee leaf illnesses in the real world. With a Macro average accuracy of 89.54%, a Weighted average precision of 90.16%, and a Micro average precision of 90.18%, the Result Analysis Using Federated Averaging CNN shows encouraging statistical findings. The recall rates varied from 88.66% for Client 2 to 93.46% for Client 4, demonstrating the model's skill in spotting the presence of sickness. F1-scores showing a harmonious balance between recall and accuracy showed values ranging from 90.99% for Client 6 to 93.35% for Client 4. A crucial statistic, accuracy, showed resilience with scores ranging from 0.95 to 0.97 for all customers. These solid statistical results demonstrate the effectiveness of the federated learning strategy and represent a significant advance in developing creative, privacy-preserving, and saleable solutions to agricultural problems. The model's ability to distinguish between different degrees of coffee leaf disease severity establishes a standard for further studies, opens the door for in-the-moment, practical applications, and opens up new possibilities for collaborative learning approaches. As a result, the federated learning CNN model's performance in this study has produced a valuable tool for battling coffee leaf diseases and paved the way for additional investigation into the federated learning and agricultural science nexus. A new age of technological empowerment, supporting sustainable practices, and fostering global food security is heralded by the merging of advanced learning methodologies with real-world agricultural requirements.

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